

The Thomson Polygon from Its Cartesian Hessian: Exact Fourier Blocks and the Sharp Linear Stability Threshold

HCS Research Program

2 September 2026

Abstract

For identical positive point vortices on a planar regular N -gon, we derive the rotating relative equilibrium directly from the logarithmic Hamiltonian and reduce its raw Cartesian Hessian by a vertexwise radial–tangential discrete Fourier transform. If $c = \Gamma/(4\pi R^2)$ and $q_m = m(N - m)$, the m th Hamiltonian block is

$$c \begin{pmatrix} 0 & q_m \\ -[2(N - 1) - q_m] & 0 \end{pmatrix}.$$

Root-of-unity orthogonality proves this formula for every $N \geq 3$. Consequently the symmetry-reduced polygon is linearly elliptic for $3 \leq N \leq 6$, linearly degenerate only in the conjugate modes $m = 3, 4$ when $N = 7$, and hyperbolic for every $N \geq 8$. We separate rotation, scale, translations, and the centered first-harmonic complement, and close the Γ , R , collision, and low- N faces. The heptagon statement is strictly linear. The result is a self-contained reconstruction of a classical ownner theorem, not a claim of literature priority or nonlinear heptagon stability.

1 Frozen Hamiltonian and relative equilibrium

Let $z_j = (x_j, y_j) \in \mathbb{R}^2$ and give every vortex the same circulation $\Gamma > 0$. We freeze the convention

$$H(z) = -\frac{\Gamma^2}{2\pi} \sum_{j < k} \log |z_j - z_k|, \quad \Gamma \dot{z}_j = \mathbf{J} \nabla_j H, \quad \mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (1)$$

Thus $-\mathbf{J}$ generates counterclockwise rotation. Take

$$a_j = R(\cos \theta_j, \sin \theta_j), \quad \theta_j = \frac{2\pi j}{N}, \quad N \geq 3, \quad R > 0. \quad (2)$$

Thomson [1] and Havelock [2] are cited as classical owners of the polygon and its linear-stability question. Cabral and Schmidt [3] provide later stability context, and Celli, Lacombe, and Pérez-Chavela [4] document polygonal relative-equilibrium lineage. No calculation or proof below is outsourced to these sources, and no literature-priority claim is made.

Lemma 1 (Angular velocity). *The configuration (2) is a relative equilibrium with*

$$\Omega = \frac{\Gamma(N - 1)}{4\pi R^2}. \quad (3)$$

For $G = H + (\Gamma\Omega/2) \sum_j |z_j|^2$, one has $\nabla G(a) = 0$.

Proof. Differentiating (1) gives

$$\nabla_j H = -\frac{\Gamma^2}{2\pi} \sum_{k \neq j} \frac{z_j - z_k}{|z_j - z_k|^2}.$$

At $j = 0$, each summand inside the sum has radial component $1/(2R)$; tangential components cancel between k and $N - k$. Rotational covariance therefore gives

$$\sum_{k \neq j} \frac{a_j - a_k}{|a_j - a_k|^2} = \frac{N - 1}{2R^2} a_j.$$

Hence $\nabla_j H(a) = -\Gamma\Omega a_j$ with (3), proving both claims and the sign of rotation under (1). $\square$

The rotating-frame linearization is

$$\mathcal{L} = \mathbf{J}_N \Gamma^{-1} D^2 G(a), \quad \mathbf{J}_N = \text{diag}(\mathbf{J}, \dots, \mathbf{J}). \quad (4)$$

All stability statements below refer to (4); they do not silently change the Hamiltonian or clock.

2 Raw Cartesian Hessian and exact Fourier blocks

For $d \neq 0$, put

$$A(d) = \frac{1}{|d|^2} - \frac{2dd^T}{|d|^4}.$$

The pairwise Cartesian blocks are

$$D_{jj}^2 H = -\frac{\Gamma^2}{2\pi} A(d_{jk}), \quad D_{jk}^2 H = +\frac{\Gamma^2}{2\pi} A(d_{jk}), \quad (5)$$

and the augmentation adds $\Gamma \Omega \mathbf{l}$ to each diagonal block. Equation (5), rather than the final closed formula, is the starting point of the independent checker.

Let $\mathbf{Q}_j$ have the radial and counterclockwise tangential unit vectors at a_j as columns. Set

$$c = \frac{\Gamma}{4\pi R^2}, \quad q_m = m(N - m), \quad \sigma_m = 2(N - 1) - q_m. \quad (6)$$

Theorem 1 (Complete DFT block). *In the local radial–tangential DFT, for every $m = 0, \dots, N - 1$,*

$$\widehat{\Gamma^{-1} D^2 G}_m = c \text{diag}(\sigma_m, q_m), \quad (7)$$

$$\mathcal{L}_m = c \begin{pmatrix} 0 & q_m \\ -\sigma_m & 0 \end{pmatrix}, \quad \mathcal{L}_m^2 = -c^2 q_m \sigma_m \mathbf{l}. \quad (8)$$

Proof. For $k \neq 0$, let $\theta = \theta_k$ and $d = a_0 - a_k$. Since $|d|^2 = 4R^2 \sin^2(\theta/2)$, substitution in $A(d)$ gives

$$A(d) = \frac{1}{4R^2 \sin^2(\theta/2)} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}.$$

Multiplication by $\mathbf{Q}_k$ turns the displayed matrix into $\text{diag}(1, -1)$. Thus the local off-diagonal block in the first block row is

$$B_k = \frac{c}{1 - \cos \theta_k} \text{diag}(1, -1).$$

If $A_N = \sum_{k=1}^{N-1} (1 - \cos \theta_k)^{-1}$, the raw diagonal block is $B_0 = c \text{diag}(2(N - 1) - A_N, A_N)$. Pairing k with $N - k$ cancels the sine parts of the DFT and yields

$$\widehat{\Gamma^{-1} D^2 G}_m = c \text{diag}(2(N - 1) - S_m, S_m), \quad S_m = \sum_{k=1}^{N-1} \frac{1 - \cos(m\theta_k)}{1 - \cos \theta_k}. \quad (9)$$

For $\zeta = e^{2\pi i/N}$,

$$\frac{1 - \cos(m\theta)}{1 - \cos \theta} = \left| \sum_{r=0}^{m-1} e^{ir\theta} \right|^2.$$

Root-of-unity orthogonality therefore gives

$$\sum_{k=0}^{N-1} \left| \sum_{r=0}^{m-1} \zeta^{kr} \right|^2 = Nm.$$

The $k = 0$ term is m^2 , so $S_m = Nm - m^2 = q_m$. This proves (7). The local rotations preserve $\mathbf{J}$; multiplying by $\mathbf{J}$ and squaring proves (8). $\square$

3 The sharp linear threshold

Theorem 2 (Thomson polygon trichotomy). *After center, angular-impulse, and rotation reduction:*

1. every block is semisimple elliptic for $3 \leq N \leq 6$;
2. for $N = 7$, exactly $m = 3, 4$ are nonzero nilpotent blocks;
3. every $N \geq 8$ has a real hyperbolic pair.

The middle statement is linear degeneracy only; it does not assert nonlinear stability or instability of the heptagon.

Proof. For $m \neq 0$, $q_m > 0$, so (8) is elliptic, nilpotent, or hyperbolic according as σ_m is positive, zero, or negative. The maximum of q_m is $\lfloor N^2/4 \rfloor$. Hence the least sign is

$$2(N-1) - \lfloor N^2/4 \rfloor.$$

It is positive for $N = 3, 4, 5, 6$ and zero at $N = 7$, where $m(7-m) = 12$ only for $m = 3, 4$. For $N = 2k \geq 8$, $k^2 - (4k-2) > 0$; for $N = 2k+1 \geq 9$, $k(k+1) - 4k = k(k-3) > 0$. Thus the maximal- q_m block is hyperbolic from eight onward. The symmetry directions removed in the statement are identified explicitly below in the revised manuscript and do not alter these signs. $\square$

N	controlling σ_m	reduced linear type	strict boundary
3–6	positive	elliptic	semisimple
7	zero at $m = 3, 4$	degenerate	no nonlinear claim
8 and above	negative for some m	hyperbolic	real pair

4 Symmetry slice and parameter faces

The full Fourier ledger prevents symmetry modes from being mislabeled as instabilities. At $m = 0$,

$$\mathcal{L}_0 = c \begin{pmatrix} 0 & 0 \\ -2(N-1) & 0 \end{pmatrix}. \quad (10)$$

Uniform tangential displacement is rotation; uniform radial displacement is its scale generalized vector. Fixing angular impulse and quotienting rotation remove (10).

The conjugate first harmonics contain the translation plane. If local coordinates are ξ_j, η_j and $w_j = \xi_j + i\eta_j$, then the center variation is

$$\delta \left(\sum_j z_j \right) = \sum_j e^{i\theta_j} w_j.$$

Fixing the center removes this complex translation plane. It does not remove the entire first-harmonic isotypic component: because $q_1 = \sigma_1 = N-1$, the Hessian there is the scalar $c(N-1)I$ and its centered complementary plane remains elliptic with frequency $c(N-1) = \Omega$. The centered, fixed-impulse rotational slice therefore removes exactly the intended Euclidean and scale directions while retaining every genuine shape degree of freedom.

The physical parameter faces are also explicit.

- $R = 0$ is a logarithmic collision and is excluded. As $R \rightarrow \infty$, all frequencies vanish like R^{-2} while the sign atlas remains fixed.
- $\Gamma = 0$ degenerates the weighted symplectic form; it is recorded only as a zero-velocity limit. Common $\Gamma < 0$ reverses time and preserves all stability signs.
- $N = 1$ is trivial. After centering, $N = 2$ has no reduced shape degree of freedom, so the theorem begins at $N = 3$.
- $N = 7, m = 3, 4$ is an exact nonzero nilpotent face. A higher-order nonlinear conclusion would require a different theorem and is not inferred.

5 Independent executable reconstruction

The finite certificate tests the derivation without pretending to prove it. The producer emits (6)–(8) as exact integer cells. The checker shares no producer function: for each $N = 3, \dots, 64$ it constructs (5) in global Cartesian coordinates,

verifies the augmented equilibrium, checks the complete $2N \times 2N$ symmetry, rotates every block by the Q_j , verifies cyclicity, performs the DFT, and applies the raw Hessian and Hamiltonian linearization to explicit rotation, scale, translation, and centered first-harmonic vectors.

SymPy separately differentiates and transforms exact raw Hessians for $N = 3, 4, 6$ and verifies the same slice actions. Its root-sum audit covers every one of the 2,077 archived mode cells by coefficient counting rather than by inserting the closed value. Two unrelated fresh output paths must reproduce the canonical JSON byte-for-byte. The checker rejects duplicate keys, nonstandard constants, unknown or missing fields, bool/type confusion, reordered or semantically duplicated rows, and all altered contract values, even when the payload hash is repaired.

6 Conclusion

The logarithmic pair Hessian and one root-of-unity sum determine every Fourier block of the equal-vortex polygon. Once rotation, scale, and translations are handled explicitly, the sign $2(N - 1) - m(N - m)$ yields the exact reduced linear transition at seven and eight vortices. The parameter ledger prevents zero circulation, collision, infinite radius, or the first harmonics from being folded into the shape theorem.

References

- [1] J. J. Thomson, *A Treatise on the Motion of Vortex Rings*, Macmillan, London (1883).
- [2] T. H. Havelock, “The stability of motion of rectilinear vortices in ring formation,” *Philosophical Magazine* **11** (70), 617–633 (1931), doi:10.1080/14786443109461714.
- [3] H. E. Cabral and D. S. Schmidt, “Stability of relative equilibria in the problem of $N + 1$ vortices,” *SIAM J. Math. Anal.* **31** (2), 231–250 (2000), doi:10.1137/S0036141098302124.
- [4] M. Celli, E. A. Lacomba, and E. Pérez-Chavela, “On polygonal relative equilibria in the N -vortex problem,” *J. Math. Phys.* **52**, 103101 (2011), doi:10.1063/1.3646115.